

Software Requirements Specification (SRS)

SmartHR + SmartResume – AI-Enhanced HR & ATS System

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional, non-functional, and system requirements for SmartHR + SmartResume, an integrated platform combining a Human Resource Management System (HRMS) with an AI-powered Applicant Tracking System (ATS). The document is intended for the development team, QA team, and the project lead to ensure a shared understanding of system expectations.

1.2 Product Overview

SmartHR + SmartResume focuses on streamlining recruitment and HR operations. The system automates resume ranking using AI, manages employee lifecycle data, attendance, salary & payroll, leave workflows, and presents key metrics through an HR dashboard.

1.3 Intended Audience

- Team Lead / Project Manager
- Backend Developers
- Frontend Developers
- QA / Test Engineers
- HR domain stakeholders

1.4 Product Objectives

- Improve hiring efficiency through automated resume screening
- Maintain complete employee records in one system
- Automate attendance, salary, and leave workflows
- Provide real-time HR insights for faster decisions

2. System Scope

SmartHR + SmartResume includes the following capabilities:

- Upload, process, and store resumes
- AI-based resume evaluation and ranking

- Complete employee management lifecycle
- Salary, payroll and salary slip generation
- Daily attendance logging
- Leave request and approval handling
- Interactive dashboard with HR insights

3. Overall Description

3.1 User Classes & Characteristics

Admin: Full access to all modules and configuration settings.

HR Manager: Manages recruitment, employee data, leave approval, attendance, and payroll.

Employee: Can apply for leave and view personal details.

3.2 Operating Environment

- Web application
- React frontend
- Node.js backend
- MongoDB database
- OpenAI API for AI scoring

3.3 Design Constraints

- Requires internet for AI screening
- Only PDF resumes supported

4. System Features

4.1 AI Resume Screening

The system automatically evaluates resumes based on HR-defined skills, identifies matched and missing skills, assigns a score, and ranks candidates.

4.2 Resume Processing (Non-AI)

Supports manual PDF uploads, text extraction, and parsing of basic details.

4.3 Employee Management

Stores complete employee profiles with CRUD operations and search/filter options.

4.4 Salary & Payroll Management

Handles salary structure, monthly salary slips, and payment tracking.

4.5 Attendance Tracking

HR marks daily attendance and the system maintains logs.

4.6 Leave Management

Employees apply for leave; HR approves or rejects requests.

4.7 HR Dashboard

Displays insights such as employee count, pending leaves, AI-ranked candidates, and payroll overview.

5. Functional Requirements

- Resume upload and extraction
- AI scoring and ranking
- Employee CRUD
- Payroll generation
- Attendance logging
- Leave application and approval
- Dashboard analytics

6. Non-Functional Requirements

- Performance: handle 500+ resumes
- AI response under 3 seconds
- Secure data access
- HR-friendly UI
- 99% uptime

7. System Architecture

Three-layer architecture: React UI → Node.js API → MongoDB. AI integrated via backend API call.

8. Data Collections

- resumes
- employees
- salaries
- attendance
- leaves

9. Conclusion

This SRS outlines the complete specification required for SmartHR + SmartResume, enabling the team lead and developers to proceed confidently with implementation.